

requests 9V, TPS54531 generates protected 5.1V.

Orange Pi and wet-side sensor 5V are switched by Q1.

USB-C PD input

VBUS TVS

PD sink 9V

3A-5A polyfuse

TPS54531 LDOA 5A buck

5A 50V Schottky

10.2k 1% 1.96k 1%

2N7002

100k

1k

100k

10nF 50V

665k

130k

37.4k

2200pF

22pF

Zero 3 26-pin

Pi Pico socket

J3

J4

J5

J6

J7

J15

J16

J8

J9

J10

J11

J12

J13

pH sensor

TDS sensor

Turbidity sensor

ORP sensor

Dissolved oxygen

NH4 ISE amp

DS18B20 water temp

BME280 air sensor

DS3231 RTC

Download button

330R if 3.3V LED

Spare analog header

TPS54531 datasheet: short SW loop, thermal copper, tight input caps.

Check pinouts against supplier datasheets before PCB order.

Review selected IC pinouts and buck layout before PCB manufacture. USB-C PD, buck regulator, Pico controller, Orange Pi header, ADCs, sensors, and RTC. Water Logger Sheet: / File: zero3-water-logger-carrier.kicad_sch		
Title: Orange Pi Zero 3 Water Logger Carrier		
Size: A3	Date: 2026-05-25	Rev: R1 schematic
KiCad E.D.A. 10.0.3		Id: 1/1